

Math 871 Topology I Worksheet 5

Product topology

1. The universal property of the product topology: given a family of topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$, endow the cartesian product

$$\prod_{\alpha \in I} X_\alpha$$

with the product topology. Given a topological space S and a family of functions $f_\alpha: Y \rightarrow X_\alpha$, consider the function

$$\begin{aligned} f: S &\longrightarrow \prod_{\alpha \in I} X_\alpha \\ y &\longmapsto (f_\alpha(y))_{\alpha \in I}. \end{aligned}$$

Show that f is continuous if and only if f_α is continuous for all $\alpha \in I$.

2. Consider the diagonal function

$$\begin{aligned} \Delta: \mathbb{R} &\longrightarrow \mathbb{R}^\omega \\ t &\longmapsto (t) = (t, t, t, \dots). \end{aligned}$$

Show that Δ is continuous when $\mathbb{R}^\omega$ is given the product topology, but not under the box topology.

3. Let $\{X_\alpha\}_{\alpha \in I}$ be any collection of topological spaces and endow

$$\prod_{\alpha \in I} X_\alpha$$

with the product topology. Let A_α be a subset of X_α for each $\alpha \in I$.

- (a) Show that if every X_α is Hausdorff, then the product $\prod_{\alpha \in I} X_\alpha$ is Hausdorff.
 (b) Show that if all the A_α are closed, then their product is closed in the product topology.
 (c) Show that

$$\prod_{\alpha \in I} \overline{A_\alpha} = \overline{\prod_{\alpha \in I} A_\alpha}.$$

- (d) Consider

$$A = \prod_{\alpha \in I} A_\alpha.$$

Show that the product topology on A induced by giving every A_α the subspace topology agrees with the subspace topology of A as a subspace of $\prod_{\alpha} X_\alpha$ with the product topology.

Continuous functions

4. Given a function $f: X \rightarrow Y$, show that the following properties hold:

- (a) For all subsets $A \subseteq X$, $A \subseteq f^{-1}(f(A))$, and equality holds if f is injective.
- (b) For all subsets $B \subseteq Y$, $f(f^{-1}(B)) \subseteq B$, and equality holds if f is surjective.
- (c) For all subsets $B_1 \subseteq B_2$ of Y , $f^{-1}(B_1) \subseteq f^{-1}(B_2)$.
- (d) Given any collection $\{B_\alpha\}_{\alpha \in I}$ of subsets of Y ,

$$f^{-1} \left(\bigcup_{\alpha \in I} B_\alpha \right) = \bigcup_{\alpha \in I} f^{-1}(B_\alpha).$$

- (e) Given any collection $\{B_\alpha\}_{\alpha \in I}$ of subsets of Y ,

$$f^{-1} \left(\bigcap_{\alpha \in I} B_\alpha \right) = \bigcap_{\alpha \in I} f^{-1}(B_\alpha).$$

- (f) For all subsets $B_1 \subseteq B_2$ of Y , $f^{-1}(B_2 \setminus B_1) \subseteq f^{-1}(B_2) \setminus f^{-1}(B_1)$.
- (g) For all subsets $A_1 \subseteq A_2$ of X , $f(A_1) \subseteq f(A_2)$.
- (h) Given any collection of subsets $\{A_\alpha\}_{\alpha \in I}$ of subsets of X ,

$$f \left(\bigcup_{\alpha \in I} A_\alpha \right) = \bigcup_{\alpha \in I} f(A_\alpha).$$

- (i) Given any collection $\{A_\alpha\}_{\alpha \in I}$ of subsets of X ,

$$f \left(\bigcap_{\alpha \in I} A_\alpha \right) \subseteq \bigcap_{\alpha \in I} f(A_\alpha)$$

with equality if f is injective.

- (j) For all subsets $A_1 \subseteq A_2$ of X , $f(A_2) \setminus f(A_1) \subseteq f(A_2 \setminus A_1)$, with equality if f is injective.

5. Give examples of:

- (a) A function $f: X \rightarrow Y$ and a collection of subsets $\{A_\alpha\}_{\alpha \in I}$ of X such that

$$f \left(\bigcap_{\alpha \in I} A_\alpha \right) \neq \bigcap_{\alpha \in I} f(A_\alpha).$$

- (b) A function $f: X \rightarrow Y$ and subsets $A_1 \subseteq A_2$ of X such that

$$f(A_2) \setminus f(A_1) \neq f(A_2 \setminus A_1).$$

6. Consider $\mathbb{R}$ under the standard topology. Give an example of a function $f: \mathbb{R} \rightarrow \mathbb{R}$ and a countable collection of closed subsets $\{A_\alpha\}_{\alpha \in I}$ such that the following all hold:

- The restrictions $f|_{A_\alpha}$ are continuous for all $\alpha \in I$.
- The closed sets A_α cover $\mathbb{R}$, meaning $\bigcup_{\alpha \in I} A_\alpha = \mathbb{R}$.
- The function f is not continuous.

7. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies on X . Show that the identity function

$$\text{id}: (X, \mathcal{T}_1) \rightarrow (X, \mathcal{T}_2)$$

is continuous if and only if $\mathcal{T}_1$ is finer than $\mathcal{T}_2$.

8. Endow $\mathbb{R}$ with the standard topology and fix real numbers $a < b$. Show that the subspaces (a, b) and $(0, 1)$ are homeomorphic and that the subspaces $[a, b]$ and $[0, 1]$ are homeomorphic.